

Patterns

Functional

-  656,250 received data and 656,250 transmitted data

Stuck at

-  33 patterns (2170 test cycles)

Transition delay

-  37 patterns (2504 test cycles)

Path delay

-  6 patterns (434 test cycles)

RS232-T2000

🔑 Trojan Description

- 🔑 Trojan trigger is a sequential comparator. Trigger input probability is $8.45e-7$ while it is externally controlled. Whenever Trojan gets triggered, its payload gains control over xmit_doneH and rec_readyH primary output signals.

🔑 Trojan Taxonomy

- 🔑 Insertion phase: Design
- 🔑 Abstraction level: Layout-level
- 🔑 Activation mechanism: Externally conditionally triggered
- 🔑 Effects: Change functionality
- 🔑 Physical characteristics: Functional

```
AND2X1 U309 (.A(sys_clk), .B(ena), .Y(sys_clk_temp));
SDDFSRX1 iDatsend_reg (.CK(sys_clk_temp), .D(iRECEIVER_rec_datSyncH_temp), .RN(1'b1), .SE(test_se), .SI(rec_dataH_rec[7]), .SN(n266), .Q(n400));
AND2X1 U308 (.A(ena), .B(iRECEIVER_rec_datSyncH), .Y(iRECEIVER_rec_datSyncH_temp));
NAND4X1 U292 (.A(ena), .B(iXMIT_state_0_), .C(iXMIT_state_1_), .D(iXMIT_state_2_), .Y(iXMIT_state_CTRL));
NAND4X1 U293 (.A(ena), .B(iXMIT_N45), .C(iXMIT_N44), .D(iXMIT_N29), .Y(iXMIT_N_CTRL_1_));
NAND4X1 U294 (.A(ena), .B(iXMIT_N27), .C(iXMIT_N26), .D(iXMIT_N25), .Y(iXMIT_N_CTRL_2_));
NAND4X1 U295 (.A(ena), .B(iXMIT_xmit_ShiftRegH_7_), .C(iXMIT_xmit_ShiftRegH_6_), .D(iXMIT_xmit_ShiftRegH_5_), .Y(iXMIT_xmit_CTRL));
OR4X1 U296 (.A(iXMIT_state_CTRL), .B(iXMIT_N_CTRL_1_), .C(iXMIT_N_CTRL_2_), .D(iXMIT_xmit_CTRL), .Y(iXMIT_CTRL));
OR2X1 U302 (.A(iXMIT_CTRL), .B(n400), .Y(iCTRL));
AND2X1 U303 (.A(iCTRL), .B(xmit_doneH_temp), .Y(xmit_doneH));
AND2X1 U401 (.A(iCTRL), .B(rec_readyH_temp), .Y(rec_readyH));
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RS232-T2000

Functional

Trojan activation probability	The number of Transmitted	The number of Received	Trojan activation	Payload Change	Simulation Length (No. Clock)	Length of each Receive/Transmission
8.45e-7	656 250	656 250	1	0	168 000 000	256/256

Structural

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Stuck-at unmatched
67 200	69 454	5 099	8	0	738	0

Transition delay

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Transition unmatched
87 811	90 736	6 767	17	0	948	0

Path delay

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Transition unmatched
34 573	35 723	2 979	2	0	380	0